CBSE Class 10 Mathematics - Answer Key

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1. **1.** Find the sum of the first 15 terms of the AP $20, 24, 28, \ldots$.

   Answer: The required sum is 720.

   Solution:

   1. Here $a = 20$, $d = 4$ and $n = 15$.
   2. Use $\frac{n}{2}[2a + (n-1)d]$ to calculate the sum.
   3. $S_{15} = \frac{15}{2}[40 + 14 \times 4] = 720$.

   Derivation:

   $$S_n = \frac{n}{2}[2a + (n-1)d]S_{15} = \frac{15}{2}[2(20) + (15-1)4]S_{15} = 720$$

   Arithmetic Progressions — sum of n terms of an AP

2. **2.** Solve the quadratic equation $x^2 - 5x + 6 = 0$ and verify the roots.

   Answer: The roots are $x = 2$ and $x = 3$.

   Solution:

   1. Factorise the polynomial: $x^2 - 5x + 6 = (x-2)(x-3)$.
   2. Set each factor equal to zero.
   3. Therefore $x = 2$ or $x = 3$.

   Derivation:

   $$x^2 - 5x + 6 = 0(x-2)(x-3) = 0 = 2, 3$$

   Quadratic Equations — factorisation method